Answer **all** the questions in the spaces provided.

1 **(a)** State the conditions required for a body to be in equilibrium.

The sum of forces acting on the body in any direction must be zero.
(The resultant force on the body is zero / No resultant force)

The sum of the moments of the forces about any point must be zero.
(The resultant moment (or torque) on the body about any axis is zero. / No resultant torque)

[2]

(b) Block A of mass 1.7 kg is connected by a light inextensible string to a counterweight B of mass *M* via a system of two massless and frictionless pulleys P and Q. Block A is balanced on a smooth plane inclined at 45°.

Pulley Q is a double pulley made up of two fixed wheels of different radii attached to a common axle, with the radii of the smaller and larger wheels *r* and 3*r* respectively as shown in Fig. 1.1.

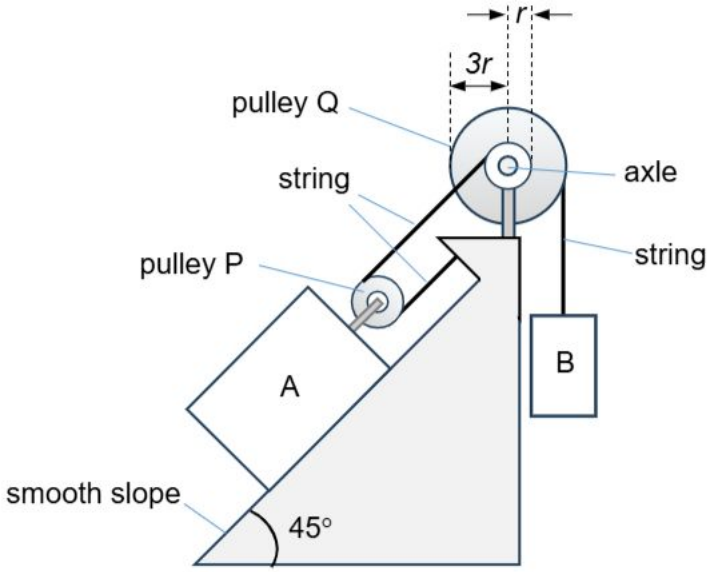
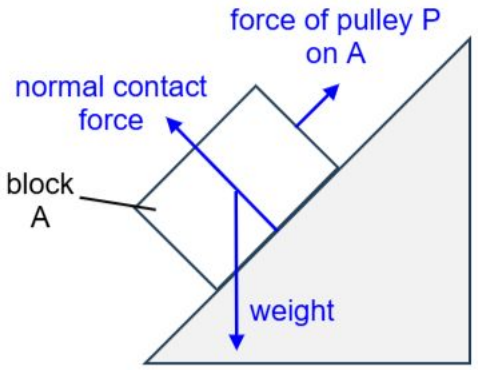


Fig.1.1

(i) Block A is in equilibrium. In Fig. 1.2, draw and label all forces acting on block A.



[B1] 3 correct forces
[B1] correct labels

Fig.1.2

[2]

(ii) Calculate the tension in the string connected to pulley P.

Block A is in equilibrium: $\sum F_{// \text{ slope}} = 0$

$m_A g \sin 45° = 2T$

$$T = \frac{m_A g \sin 45°}{2} = \frac{(1.7)(9.81)\sin 45°}{2} \quad \text{[M1]}$$

$$= 5.90 \text{ N} \quad \text{[A1]}$$

tension = 5.90 N N [2]

(iii) Determine the mass M of counterweight B.

Taking pivot about axel, $\sum \tau = 0$

CW moments = ACW moments

$m_B g(3r) = Tr$

$$m_B = \frac{Tr}{g(3r)} = \frac{T}{3g} \quad \frac{5.90}{(3)(9.81)} \quad \text{[M1]}$$

$$= 0.200 \text{ N} \quad \text{[A1]}$$

M = 0.200 kg kg [2]

(iv) Suggest and explain a change that can be made to the pulley such that block A will accelerate up the slope.

The larger wheel of the double pulley can be made larger with the smaller wheel remaining constant / more than 3 times larger than the smaller wheel. [M1]

This is so that the clockwise moment will be larger than the anticlockwise moment / net clockwise torque causing the block A to slide up the slope. [A1] [2]

[Total: 10]